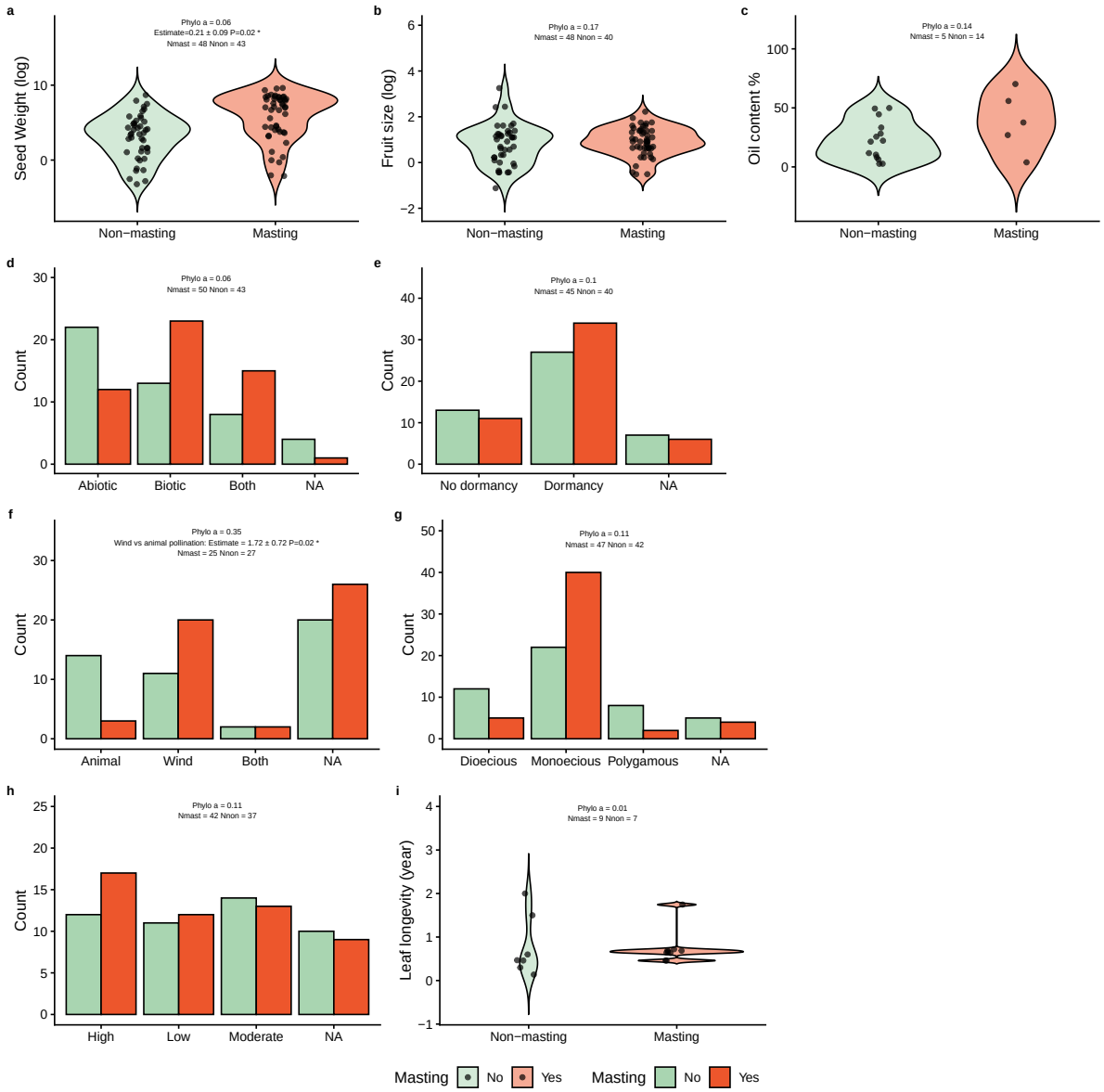

Figure 1: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for angiosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter (α), and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

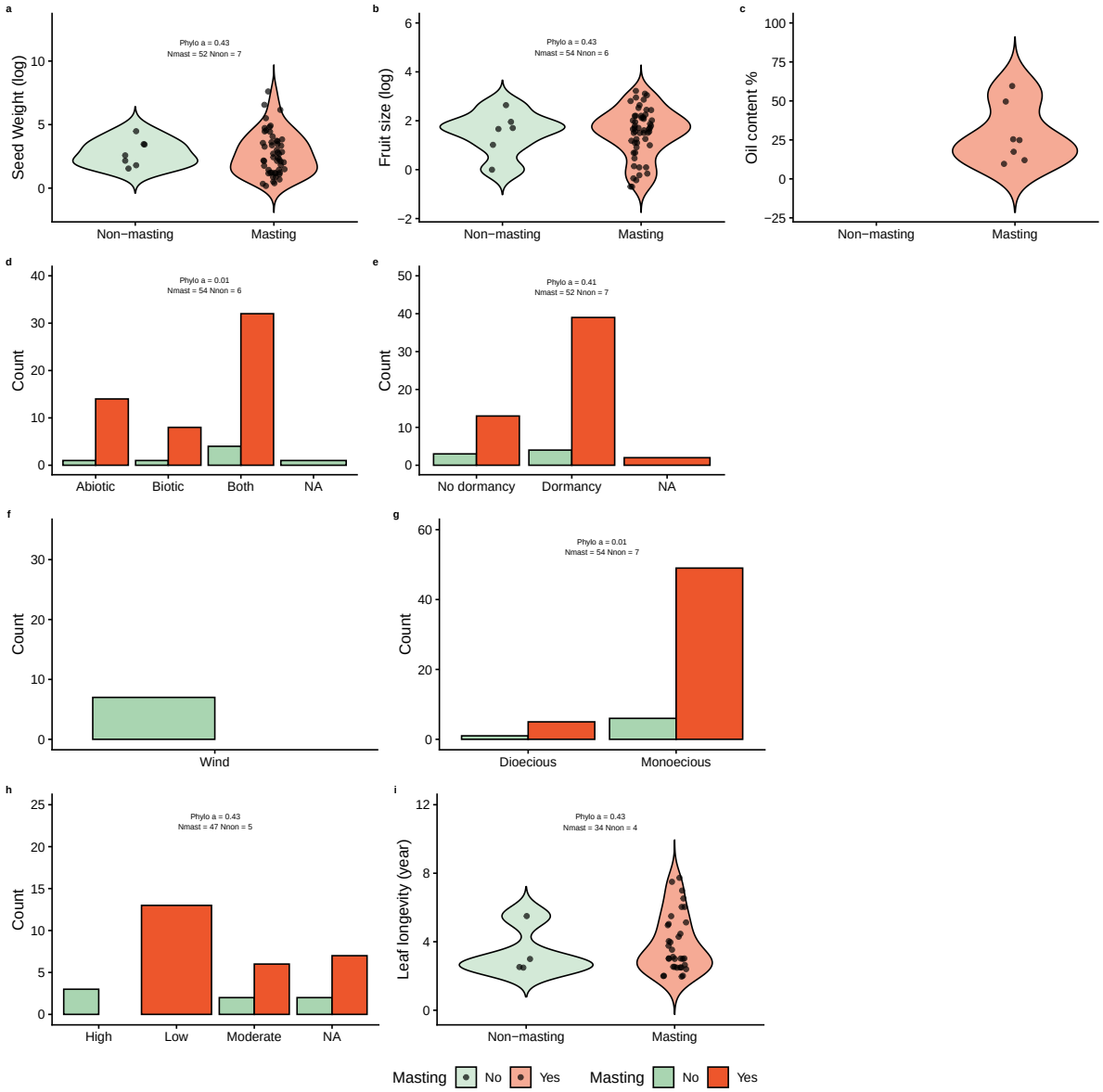

Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for gymnosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter (α), and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

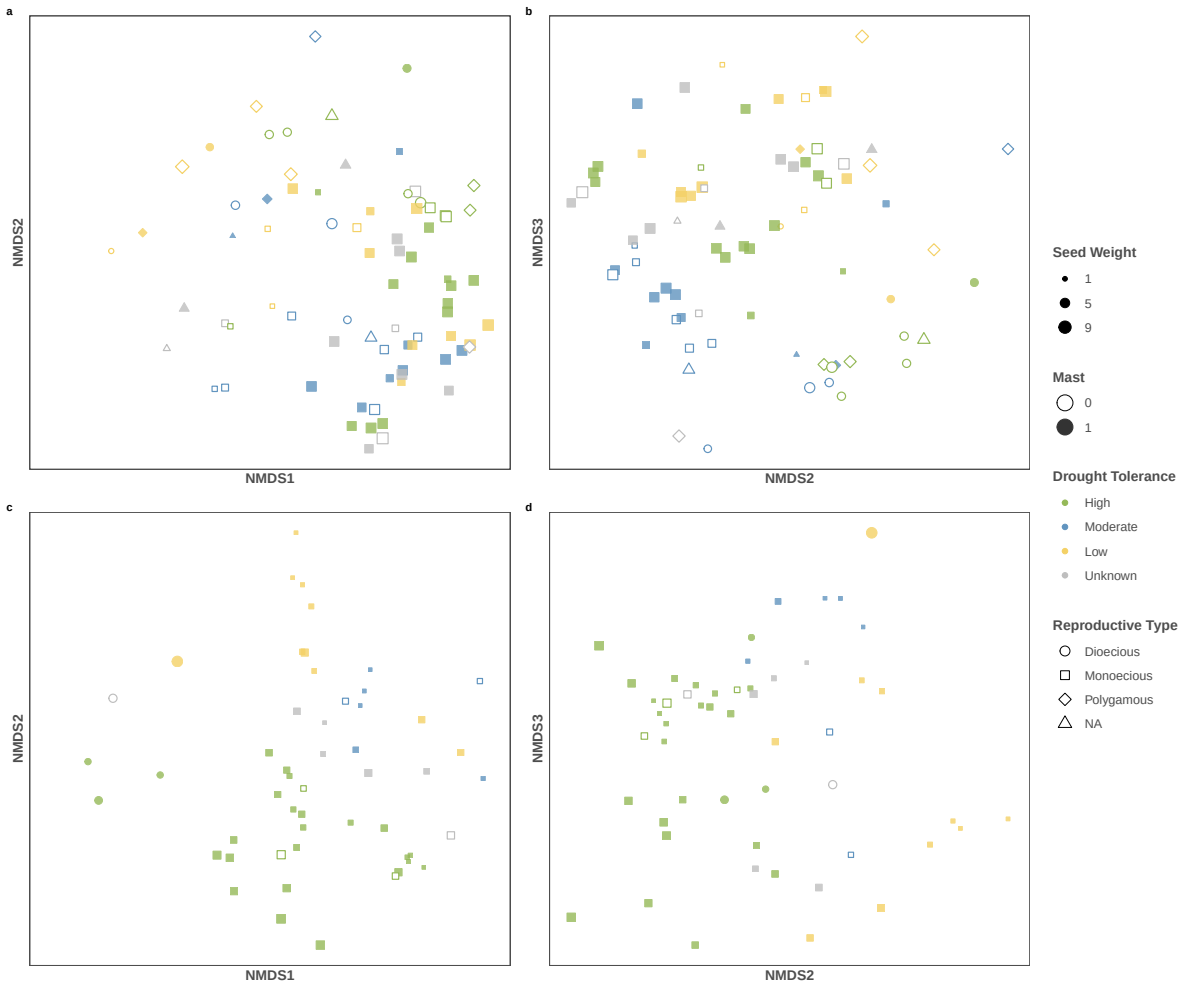

Figure 3: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and conifer species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) conifer NMDS1–NMDS2, and (d) conifer NMDS2–NMDS3.

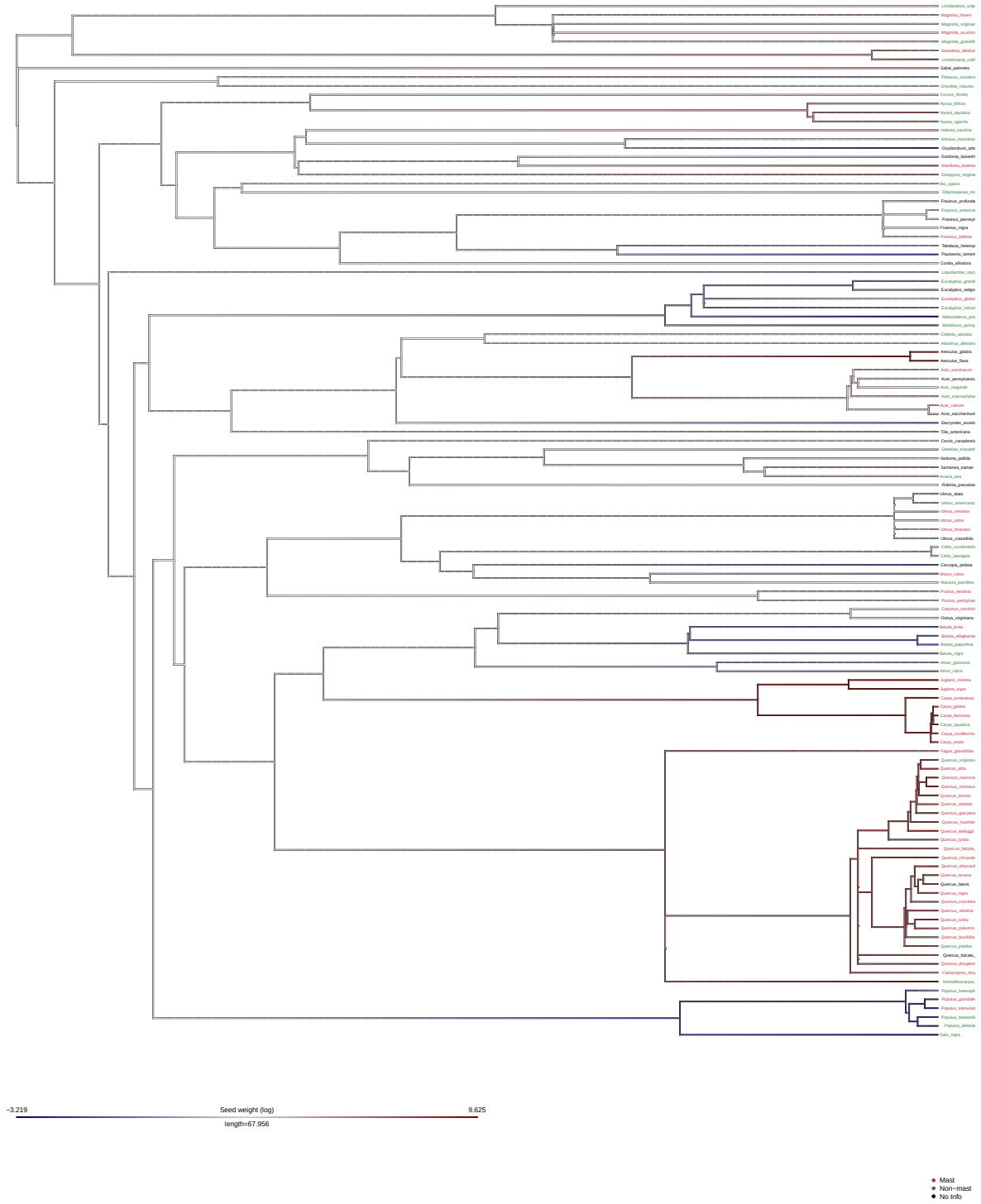

Figure 4: Phylogenetic tree for all species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by masting or not. We calculated the Blomberg's K for seed weight.

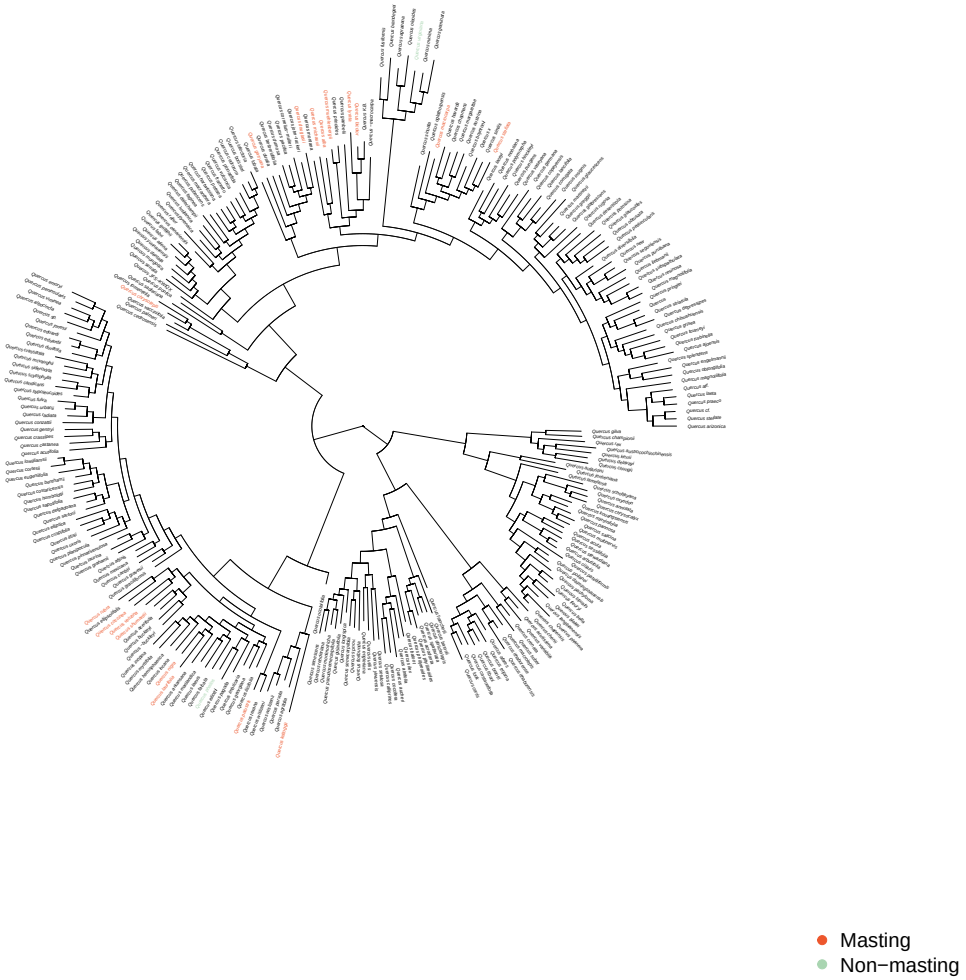

Figure 5: Phylogenetic tree for all Quercus species, Quercus species included in this study is color coded.

Group	Trait	Predictor	Estimate	SE	P	Phylo α	N
Gymnosperm	Seed dispersal	Biotic vs abiotic	-0.83	0.81	0.30	0.01	60
Gymnosperm	Seed dispersal	Both vs abiotic	-1.11	0.67	0.10	0.01	60
Gymnosperm	Seed dormancy	Dormant vs non-dormant	0.71	0.70	0.32	0.41	59
Gymnosperm	Reproductive type	Monoecious vs dioecious	-1.14	1.02	0.26	0.01	61
Gymnosperm	Seed weight (log)	logSeedWeight	-0.03	0.25	0.91	0.43	59
Gymnosperm	Fruit size (log)	logFruit	0.03	0.36	0.94	0.43	60
Gymnosperm	Seed size (log)	logSeedSize	-0.37	0.93	0.69	0.41	51
Gymnosperm	Leaf longevity (years)	Leaf longevity (years)	0.15	0.34	0.67	0.43	38
Gymnosperm	Drought tolerance	Low vs high drought tolerance	0.91	1.64	0.58	0.43	52
Gymnosperm	Drought tolerance	Moderate vs high drought tolerance	-1.56	1.06	0.14	0.43	52
Angiosperm	Dispersal mode	Biotic vs abiotic	0.55	0.53	0.30	0.06	93
Angiosperm	Dispersal mode	Both vs abiotic	0.55	0.57	0.33	0.06	93
Angiosperm	Pollination mode	Wind vs animal pollination	1.72	0.72	0.02	0.35	52
Angiosperm	Pollination mode	Wind vs animal pollination and animals	1.23	1.15	0.29	0.35	52
Angiosperm	Seed dormancy	Dormant vs non-dormant	0.03	0.50	0.95	0.10	85
Angiosperm	Reproductive type	Monoecious vs dioecious	0.90	0.68	0.18	0.11	89
Angiosperm	Reproductive type	Monoecious vs polygamous	-0.13	0.96	0.89	0.11	89
Angiosperm	Seed weight (log)	logSeedWeight	0.21	0.09	0.02	0.06	91
Angiosperm	Fruit size (log)	logFruit	0.08	0.30	0.80	0.17	88
Angiosperm	Seed size (log)	logSeedSize	0.24	0.30	0.43	0.08	69
Angiosperm	Oil content (%)	Oil content (%)	0.04	0.03	0.20	0.14	19
Angiosperm	Leaf longevity (years)	Leaf longevity (years)	0.26	0.95	0.78	0.01	16
Angiosperm	Drought tolerance	Low vs high drought tolerance	0.04	0.48	0.92	0.11	79
Angiosperm	Drought tolerance	Moderate vs high drought tolerance	-0.13	0.46	0.79	0.11	79

Table 1: Phylogenetic Generalized Linear Model Results

Group	Trait	λ
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Seed size (Log)	0.96
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Seed dormancy	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	0.00
Angiosperm	Drought tolerance	1.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Seed size (Log)	0.26
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Seed dormancy	1.00
Gymnosperm	Reproductive type	1.00
Gymnosperm	Drought tolerance	1.00

Table 2: Phylogenetic Signal (Pagel's λ) for All Traits